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MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

JEE Main Topic

(English Medium)

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PRINCIPLE OF MATHEMATICAL INDUCTION

1. THEOREM-I

If P(n) is a statement depending upon n, then to prove P(n) by induction, we procced as follows :

(i) Verify the validity of P(n) for n = 1

(ii) Assume that P(n) is true for any positive integer m and then using it establish the validity of P(n) for n = m + 1.

Then P(n) is true for each $n \in N$

2. THEOREM-II

If P(n) is a statement depending upon n but begining with any positive integer k, then to prove P(n) by Induction, we procced as follows :

(i) Verify the validity of P(n) for n = k.

(ii) Assume that the P(n) is true for $n = m \geq k$.

Then using it estabish the validity of P(n) for n = m + 1.

Then P(n) is true for each $n \geq k$

3. SOME USEFUL RESULT BASED ON PRINCIPLE OF MATHEMATICAL INDUCTION :

For any natural number n

(i) $1 + 2 + 3 + ... + n = \Sigma n = \frac{n(n+1)}{2}$

(ii) $1^2 + 2^2 + 3^2 + ... + n^2 = \Sigma n^2 = \frac{n(n+1)(2n+1)}{6}$

(iii) $1^3 + 2^3 + 3^3 + ... + n^3 = \Sigma n^3 = (\Sigma n)^2 = \left\{\frac{n(n+1)}{2}\right\}^2$

(iv) $2 + 4 + 6 + ... + 2n = \Sigma 2n = n(n + 1)$

(v) $1 + 3 + 5 + ... + (2n - 1) = \Sigma(2n - 1) = n^2$

(vi) $x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + ... + xy^{n-2} + y^{n-1})$

(vii) $x^n + y^n = (x + y)(x^{n-1} - x^{n-2}y + x^{n-3}y^2 + ... - xy^{n-2} + y^{n-1})$

when n is odd positive integer

4. IMPORTANT TIPS :

(i) Product of r successive integers is divisible by r!

(ii) For $x \neq y$, $x^n - y^n$ is divisible by

(a) x + y, if n is even (b) x – y, if n is even or odd

(iii) $x^n + y^n$ is divisible by

x + y, If n is odd

(iv) For solving objective question related to natural numbers we find out the correct alternative by negative examination of this principle. If the given statement is P(n), then by putting n = 1, 2, 3 .... in P(n) we decide the correct answer. We also use the above formulae established by this principle to find the sum of n terms of a given series. For this we first express T_n as a polynomial in n and then for finding S_n, we put Σ before each term of this polynomial and then use above results of Σn, Σn^2, Σn^3 etc.

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SOLVED EXAMPLES

Ex.1 Use the principle of mathematical induction to show that $5^{2n+1} + 3^{n+2}.2^{n-1}$ divisible by 19 for all natural numbers n.

Sol. Let $P(n) = 5^{2n+1} + 3^{n+2}.2^{n-1}$

Step I : For n = 1

$P(1) = 5^{2+1} + 3^{1+2}.2^{1-1}$

$= 125 + 27$

$= 152$, which is divisible by 19.

Therefore, the result is true for n = 1.

Step II : Assume that the result is true for n = k, i.e. $P(k) = 5^{2k+1} + 3^{k+2}$, 2^{k-1} is divisible by 19.

$\Rightarrow P(k) = 19r$, where r is an integer.

Step III : For n = k + 1

$P(k + 1) = 5^{2(k+1)+1} + 3^{k+1+2}. 2^{k+1-1}$

$= 5^{2k+3} + 3^{k+3}.2^{k}$

$= 25.5^{2k+1} + 3.3^{k+2}.2.2^{k-1}$

$= 25.5^{2k} + 6.3^{k+2}.2^{k-1}$

Now $25.5^{2k+1} + 6.3^{k+2}2^{k-1} = 25.(5^{2k+1} + 3^{k-2}.2^{k-1}) - 19.3^{k+2}.2^{k-1}$

i.e. $P(k + 1) = 25\ P(k) - 19.3^{k+2}.2^{k-1}$

But we know that P(k) is divisible by 19. Also $19.3^{k+2}.2^{k-1}$ is clearly divisible by 19.

Hence P(k + 1) is divisible by 19. This shows that the result is true for n = k + 1. Hence by the priciniple of mathematical induction, the result is true for all $n \in N$.

Ex.2. Use the principle of mathematical induction to show that $1.3 + 2.4 + \ldots\ldots + n.(n+2) = \frac{1}{6}n(n+1)(2n+7)$.

Sol. Let P(n) : $1.3+ 2.4 +\ldots..+n.(n+2) = \frac{1}{6}n(n + 1)(2n + 7)$

Step I : For n = 1

LHS of $P(1) = 1.3 = 3 = \frac{1}{6}.1.2.9 = \frac{1}{6}.1(1+1)(2.1+7) =$ RHS of P(1)

So P(1) is true

Step II : Now assume P(k) is true, for some natural number k, i.e

$1.3 + 2.4 +\ldots..+k.(k + 2) = \frac{1}{6}k(k + 1)(2k + 7)$.

Now deduce P(k + 1).

LHS of $P(k + 1) = 1.3 + 2.4+\ldots\ldots+k.(k+2) + (k + 1).(k + 1 + 2)$

= (LHS of P(k)) + (k + 1) (k + 3)

= (RHS of P(k)) + (k + 1) (k + 3), (by inductive assumption)

$= \frac{1}{6}k(k+1)(2k+7)+(k+1)(k+3)$

$= \frac{1}{6}(k+1)\left(k(2k+7)+6(k+3)\right)$

$= \frac{1}{6}(k+1)\left(2k^2+13k+18\right)$

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$$= \frac{1}{6}(k+1)(k+2)(2k+9)$$

$$= \frac{1}{6}(k+1)(k+1+1)\left(2(k+1)+7\right)$$

= RHS of P(k + 1).

So P(k + 1) is true, if P(k) is true.

Hence by induction P(n) is true for all natural numbers n.

Ex.3 Use the principle of mathematical induction to show that for any positive integer number n, $n^3 + 2n$, is divisible by 3.

Sol. Statement P(n) is defined by $n^3 + 2$ n is divisible 3

Step 1 : We first show that P(1) is true. Let n = 1 and calculate $n^3 + 2n$

$1^3 + 2(1) = 3$

Hence P(1) is true.

Step 2 : We now assume that P(k) is true $k^3 + 2k$ is divisible by 3. is equivalent to

$k^3 + 2k = 3M$, where M is a positive integer.

We now consider the algebraic expression $(k + 1)^3 + 2(k + 1)$; expand it and group like terms.

$(k + 1)^3 + 2(k + 1) = k^3 + 3k^2 + 5k + 3$

$= [k^3 + 2k] + [3k^2 + 3k + 3]$

$= 3M + 3[k^2 + k + 1] = 3[M + k^2 + k + 1]$

Hence $(k + 1)^3 + 2(k + 1)$ is also divisible by 3 and therefore statement P(k + 1) is true.

Ex.4 Prove that $3^n > n^2$ for n = 1, n = 2 and use the mathematical induction to prove that $3^n > n^2$ for n, a positive integer greater than 2.

Sol Statement P(n) is defined by

$3^n > n^2$

Step 1 : We first show that P(1) is true. Let n =1 and calculate 3^1 and 1^2 and compare them

$3^1 = 3$

$1^2 = 1$

3 is greater than 1 and hence P(1) is true.

Let us also show that P(2) is true.

$3^2 = 9$

$2^2 = 4$

Hence P(2) is also true.

Step 2 : We now assume that P(k) is true

$3^k > k^2$

Multiply both sides of the above inequality by 3.

$3 * 3^k > 3* k^2$

The left side is equal to 3^{k+1}. For k > 2, we can write

$k^2 > 2$ k and $k^2 > 1$

We now combine the above inequalities by adding the left hand sides and the right hand sides of the two inequalities.

$2k^2 > 2k + 1$

We now add k^2 to both sides of the above inequality to obtain the inequality

$3k^2 > k^2 + 2k + 1$

Factor the right side we can write

$3*k^2 > (k + 1)^2$

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If $3 * 3^k > 3*k^2$ and $3*k^2 > (k + 1)^2$ then

$$3 * 3^k > (k + 1)^2$$

Rewrite the left side as 3^{k+1}

$$3^{k+1} > (k + 1)^2$$

* Which proves that P(k + 1) is true.

Ex.5. Use mathematical induction to prove De Moiver's theorem

$[R (\cos t + i \sin t)]^n = R^n(\cos x\, nt + i \sin nt)$ for n a positive integer.

Sol. Step 1 : For n = 1

$[R (\cos t + i \sin t)]^1 = R^1(\cos 1.t + i \sin 1.t)$

It can be easily be seen that the two sides are equal.

Step 2 : We now assume that the theorem is true for n = k, hence $[R(\cos t + i \sin t)]^k = R^k(\cos kt + i \sin kt)$

Multiply both sides of the above equation by $R(\cos t + i \sin t)$

$[R (\cos t + i \sin t)]^k R (\cos t + i \sin t) = R^k(\cos kt + i \sin kt) R(\cos t + i \sin t)$

Rewrite the above as follows

$[R (\cos t + i \sin t)]^{k+1} = R^{k+1}[(\cos kt \cos t - \sin kt \sin t) + i(\sin kt \cos t + \cos kt \sin t)]$

Trigonometric identities can be used to write the trigonometric expressions $(\cos kt \cos t - \sin kt \sin t)$ and $(\sin kt \cos t + \cos kt \sin t)$ as follows

$(\cos kt \cos t - \sin kt \sin t) = \cos(kt + t) = \cos(k + 1)t$

$(\sin kt \cos t + \cos kt \sin t) = \sin(kt + t) = \sin(k + 1)\, t$

Substitute the above into the last equation to obtain.

$[R (\cos t + i \sin t)]^{k+1} = R^{k+1}[\cos (k + 1)\, t + \sin(k + 1)t]$

It has been established that the theorem is true for n = 1 and that if it assumed true for n = k it is true for n = k + 1.

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